DAY 10 — Exploratory Data Analysis (EDA) & Dataset Understanding

Day 10 focused on **Exploratory Data Analysis (EDA)**, one of the most important stages in any data science workflow. EDA helps understand the dataset deeply before applying machine learning algorithms. During today's session, the emphasis was on learning how to identify trends, detect anomalies, observe relationships between variables, and prepare the data for modeling.

🔹 1. Understanding the Purpose of EDA

EDA helps answer questions such as:

- What does the dataset contain?
- Are there missing values?
- What patterns exist in the data?
- Are there any outliers?
- What relationships exist between variables?
- Which features are important?

We learned that EDA is the backbone of all ML workflows because it guides what transformations, models, and preprocessing steps are required.

🔸 2. Dataset Structure Examination

We performed operations like:

- df.head() — to view the top rows
- df.tail() — to view the bottom rows
- df.info() — to check data types
- df.describe() — summary statistics
- df.isnull().sum() — missing values

These steps helped identify:

- Numerical vs categorical columns
- Continuous vs discrete values
- Data type mismatches
- Missing value patterns

🔸 3. Handling Missing Values

We explored techniques such as:

- Dropping missing rows
- Filling missing values using mean/median
- Forward-fill and backward-fill
- Interpolation for time-series data

In stock price analysis, missing values can lead to incorrect forecasting, so accuracy here was important.

🔸 4. Outlier Detection

We used:

- Boxplots
- IQR method
- Z-score detection

Outliers can distort model predictions, especially in financial datasets where extreme values may indicate sudden market events.

🔸 5. Feature Relationship Analysis

We examined:

- Correlation heatmaps
- Scatter plots
- Pair plots

These visualizations helped identify:

- Positive and negative correlations
- Redundant features
- Key predictors for future forecasting

🔸 6. Understanding Data Distribution

We plotted:

- Histograms
- KDE plots
- Distribution curves

This helped understand:

- Skewness
- Kurtosis
- Normality of data

Financial datasets are usually not normally distributed, and today's analysis confirmed this behavior.

Key Takeaways from Day 10

- EDA is essential for building high-quality ML models
- Visualizations reveal hidden patterns
- Outliers can drastically affect predictions
- Correlation analysis guides feature engineering
- Understanding distributions helps with scaling and normalization

Day 10 gave me confidence in inspecting datasets thoroughly and preparing them in a structured, meaningful way.